STA 108 Final Project

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Part I- Multiple Linear Regression I

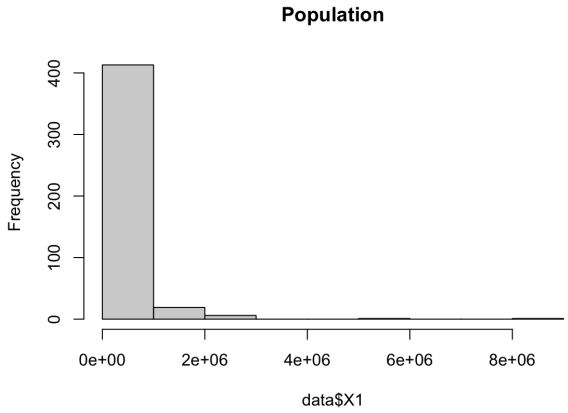
1. **Graphical summaries**

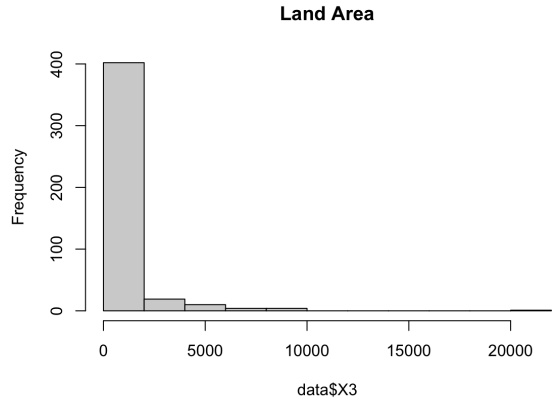
Model A: predictors X1 (Population), X3 (Land Area), X2 (Total Income)

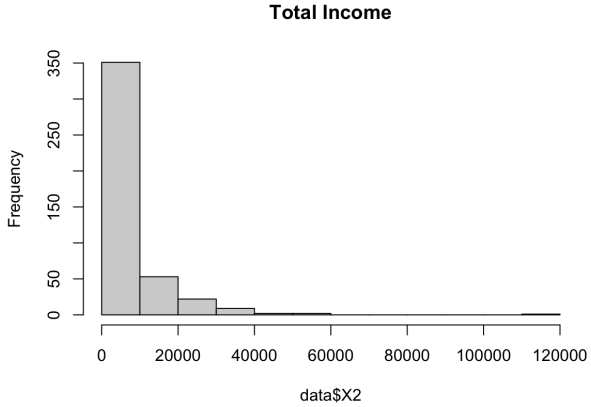
Population (X1) – The histogram is almost right-skewed.
Most counties have relatively modest populations packed near the left side of the axis, and only a few counties have very huge numbers in the millions. Also, it creates a long right tail. These huge counties could be considered strategically important locations or potential exceptions.
Land Area (X3) – The histogram is also highly shifted to the right.
Many counties have small land expanses, while only a few counties have quite large land areas (up to around 20,000). Again, such large-area counties are potential rarities. The dispersion is not typical.
Total Income (X2) – The distribution of total income is strongly shifted to the right as well.
A selected few of counties have extraordinarily high incomes (up to over 120,000), but the majority of observations are concentrated at rather average income levels. This diagram shows significant right tails and maybe exceptions.
These very strongly right-skewed distributions imply that there may be only a small group of very large counties which dominate the regression line. In practical terms, this means that transformations like taking logarithms of population, land area, or total income, or using approaches that down-weight extreme counties, may help stabilize the variance and make the relationships more linear.

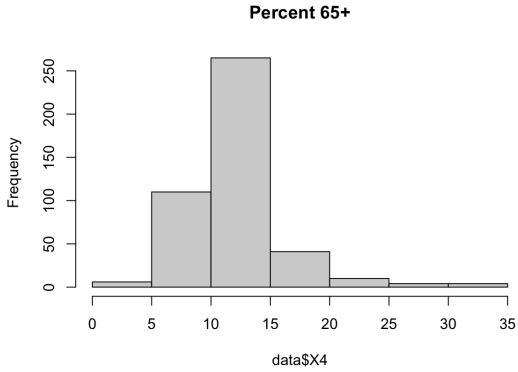
Model B: predictors Density = X1/X3, X4 (Percent 65+), X2 (Total Income)

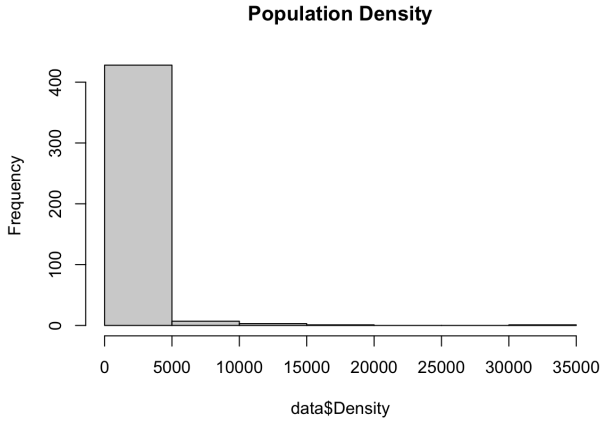
Population Density (X1/X3) – The histogram of density is again very right-skewed.
Most counties have relatively low population density, with a long list of a few highly populated counties. So the density has significant positive shifts and several extreme findings.
Percent 65+ (X4) – The "Percent 65+" histogram is much more concentrated.
Between 8% and 16% is where the majority of observations decreased. The distribution has been slightly shifted to the right because there are a few counties with very high percentages of old people (above 25–30%) and a few with very low percentages. Overall, X4 looks closer to symmetric than the other predictors.
Total Income (X2) – Same as in Model A: strongly shift to the right with a few very high-income counties.
Again, the right-skewed shapes for density and total income indicate that most counties are small or moderate, while a few extremely large or wealthy counties form long right tails. These extreme counties may act as leverage points in the regression and can slightly distort the fitted line if they are not handled carefully.



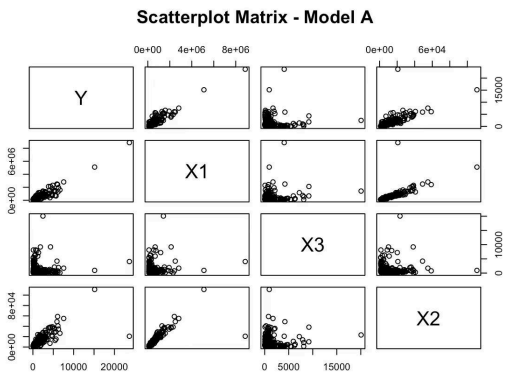


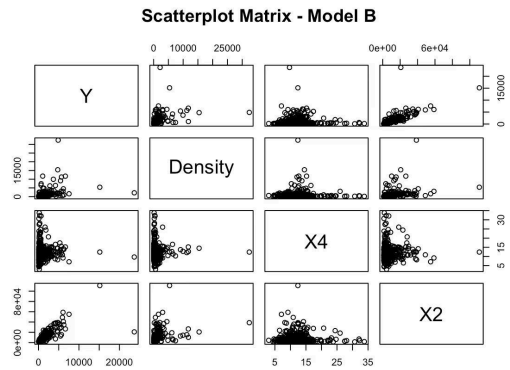






2. Scatterplot & correlation





	Y	X1	X2	X3
Y	1.00000000	0.9402486	0.9481106	0.07807466
X1	0.94024859	1.0000000	0.9867476	0.17308335
X2	0.94811057	0.9867476	1.0000000	0.12707426
X3	0.07807466	0.1730834	0.1270743	1.00000000

	Y	Density	X4	X2
Y	1.00000000	0.40643863	-0.00312863	0.94811057
Density	0.40643863	1.00000000	0.02918445	0.31620475
X4	-0.00312863	0.02918445	1.00000000	-0.02273315
X2	0.94811057	0.31620475	-0.02273315	1.00000000

Model A (Y, X1, X3, X2)

For Model A, the correlation matrix shows very strong linear relationships between Y and both Population(X1) and Total Income(X2):

Corr(Y, X1) = 0.94, Corr(Y, X2) = 0.95.
The scatterplot matrix indicates that Y increases approximately linearly with both income and population. These should be exceptionally significant positive linear correlations. Thus, population and total income are both very strong predictors of Y.
Corr(Y, X3) = 0.08, which is very close to zero.
This indicates only a very weak linear association between Y and land area. The scatterplots are also quite separate.
Corr(X1, X2) = 0.99 (0.9867).
The population and total income show strong variation. It is indicated by this extremely strong linear association. Since total income is significantly impacted by population size, including both in the same model will render their separate regression coefficients unstable and closely related.
It amounts to one case of severe multicollinearity in regression. It may cause the individual coefficients of X1 and X2 to have large standard errors, their signs and magnitudes to be sensitive to minor variation in the data or model specifications, and it may be challenging to discern the distinct contribution of each factor in the model even though the overall model continues to predict Y very well.
Corr(X1, X3) = 0.17, Corr(X2, X3) = 0.13.
These are small and very separate in scatterplots.

Model B (Y, Density, X4, X2)

The complexity is significantly lower in Model B than in Model A since all of the correlations between the predictors are smaller than roughly 0.32 in size. Y and X2 continue to have the strongest link. Also, the density has little effect, and X4 seems to be a fairly poor predictor.
Corr(Y, X2) = 0.95 again
Total income remains the strongest predictor of Y, just as in Model A.
Corr(Y, X4) ≈ −0.003, essentially zero.
Corr(Y, Density) = 0.41, indicating a moderate positive linear relationship.
Countries with higher population density tend to have larger Y, but the relationship between the two is much less than that between Y and wealth.

3. **Fitted equations & coefficient interpretation**

```
Call:
lm(formula = Y ~ X1 + X3 + X2, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-1855.6  -215.2   -74.6    79.0  3689.0 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.332e+01  3.537e+01  -0.377 0.706719    
X1           8.366e-04  2.867e-04   2.918 0.003701 ** 
X3          -6.552e-02  1.821e-02  -3.597 0.000358 ***
X2           9.413e-02  1.330e-02   7.078 5.89e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 560.4 on 436 degrees of freedom
Multiple R-squared:  0.9026,	Adjusted R-squared:  0.902 
F-statistic:  1347 on 3 and 436 DF,  p-value: < 2.2e-16
```

```
Call:
lm(formula = Y ~ Density + X4 + X2, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-3055.75  -175.30   -38.05    72.88  3045.81 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.706e+02  8.353e+01  -2.042   0.0418 *  
Density      9.616e-02  1.224e-02   7.857  3.1e-14 ***
X4           6.340e+00  6.384e+00   0.993   0.3212    
X2           1.266e-01  2.084e-03  60.723  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 533.5 on 436 degrees of freedom
Multiple R-squared:  0.9117,	Adjusted R-squared:  0.9111 
F-statistic:  1501 on 3 and 436 DF,  p-value: < 2.2e-16
```

Model A

Intercept = −13.32

When the population, land area, and total income are all zero, the predicted value of Y is −13.32.
The intercept is just a mathematical constant with minimal practical consequence because this situation is not practical for a real county.

X1 (Population) coefficient = 0.0008366

An increase of one unit in population leads an average increase of 0.0008366 units in expected Y when land area and total income are held constant.
Since the coefficient is positive and statistically significant, a larger population is associated with a larger Y after considering factors the other variables.

X3 (Land Area) coefficient = −0.06552

When population and total income are held constant, an increase of one unit in land area follows with a 0.0655-unit decrease in what is expected of Y.
The negative and significant coefficient indicates that for counties with the same population and income, larger land area corresponds to slightly smaller Y.

X2 (Total Income) coefficient = 0.09413

When population and land area are held constant, an increase of one unit in total income results in an average increase of 0.0941 units in the predicted Y.
Even after adjusting for population and location, counties with greater total incomes usually have much greater Y. This has a significant and positive effect.

Model B

Intercept = −170.6

The expected Y is -170.6 when Density, Percent 65+, and Total Income are all zero.The intercept is just a mathematical constant with no practical significance because this is again impractical.

Density coefficient = 0.09616

The expected Y increases by 0.0962 units for each unit increase in population density, while the percentage of individuals 65 and older and total income remain same.The relationship is still positive and highly statistically significant after controlling for wealth and age structure. It is suggesting that Y is generally greater in larger counties.

X4 (Percent 65+) coefficient = 6.34

Keeping density and total income constant, a one percentage point increase in the number of residents over 65 has been connected with an increase in the expected Y of about 6.34 units.
However, since the p-value is 0.32, which indicates that this effect is not statistically significant, we do not have strong evidence that the percentage of 65+ has a real linear relationship on Y.

X2 (Total Income) coefficient = 0.1266
The expected Y increases by an average of 0.1266 units for every unit rise in total income, while Density and Percent 65+ remain constant.
This positive and very significant coefficient shows once more that total income is the greatest predictor of Y in Model B.

4. Compute and Compare R2

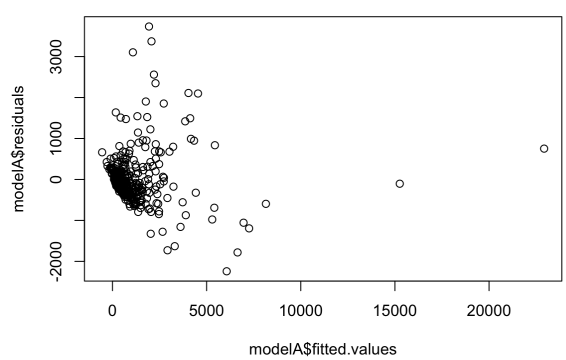
Model A:
Multiple $R_A^2 = 0.9026$
Abjusted $R_A^2 = 0.9020$
Residual standard error ≈ 560.4

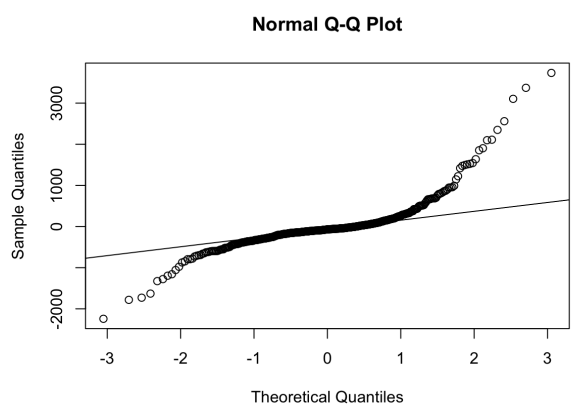
Model B:
Multiple $R_B^2 = 0.9117$
Abjusted $R_B^2 = 0.9111$
Residual standard error ≈ 533.5

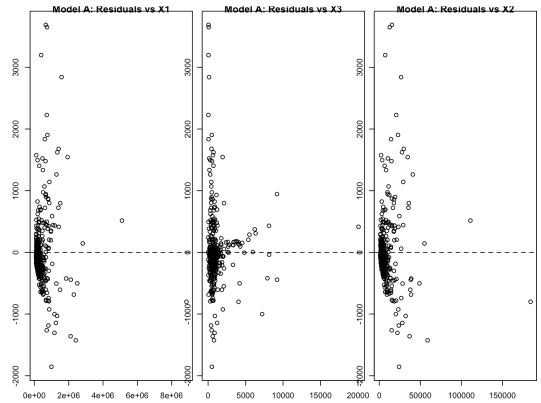
Model B is better than Model A according to R2, adjusted R2, and residual standard error. This indicates that Model B usually has smaller residuals and explains slightly greater variation in Y even after controlling for the number of predictors.

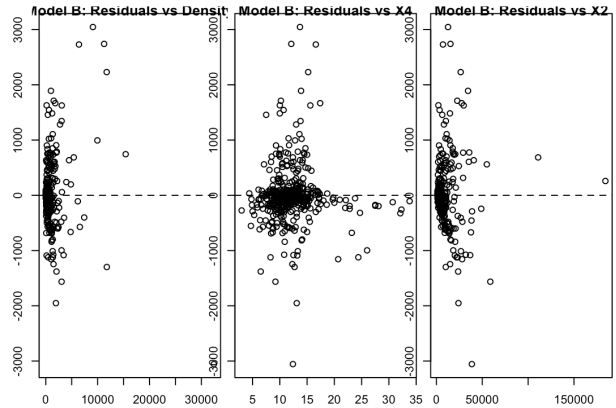
Because of this, Model B provides a total better match to the data. When combined with the relationship analysis, it shows that Model B's predictors have less interaction. Model B appears to generate more consistent and acceptable coefficient values in addition to giving a slightly superior fit.

5. Residuals & Q–Q plot









This Q-Q graphic shows the residuals from Model A. Most of the residuals are roughly normal, as evidenced by the fact that most of the spots in the middle part of the plot are rather close to the straight reference line. However, the two tails vary greatly from the line: On the left tail, the points are below the line (large negative residuals). On the right tail, the points are located much above the line (large positive residuals). The S-shaped pattern with heavy tails indicates that the error distribution is more heavy-tailed than a normal distribution, and there are greater degrees of positive and negative residuals than we would expect under absolute normally. This suggests that the standard deviation assumption is broken by the tails of Model A, based on statistical analysis. The main conclusions on general trends are probably still acceptable because the sample size is large and the center part of the distribution is nearly normal. On the other hand, tests and confidence intervals based on standard mistakes might be less accurate.

This Q-Q figure B was created using the same method. It uses the residuals from Model B. Most of the points are closer to the reference line than in Model A, especially in the central region. It is indicating that the middle half of the residual distribution is rather normal.With some of the most negative residuals going below the line and some of the most positive residuals rising above it, there are still apparent variations in the tails. Overall, it seems that both models satisfy the main presumptions needed for multiple linear regression.

6. **Add two-factor interactions**

Model A:

Multiple R^2= 0.9026
Adjusted R^2=0.9020
RSE ≈ 560.4
Multiple R^2= 0.9064
Adjusted R^2=0.9051
RSE ≈ 551.4

Model B:

Multiple $R^2 = 0.9117$
Adjusted $R^2 = 0.9111$
RSE ≈ 533.5

Multiple $R^2 = 0.9230$
Adjusted $R^2 = 0.9220$
RSE ≈ 500.0

When the three two-factor interactions are considered, the residual standard error decreases from 560.4 to 551.4 and the new R2 rises from 0.9020 to 0.9051. Fit has improved both briefly and considerably.

X1, X2, and X3 continue to be the main effects in the updated the standard model. X1:X2 is not significant (p = 0.10) among the interaction terms, whereas X1:X3 and X2:X3 are. This means that although their combined effect is less significant, the effects of population and total income on Y may differ according on the location.The impact of each predictor now depends on the values of the others. Also, it has making the model more complex and challenging to understand. In general, interactions lead to a small improvement in model fit at the expense of interpretability.

Part II — Multiple Linear Regression II

1. **Report the Value**

   These are the values of partial coefficients of determination for each single additional variable.

R^2_Y,X3|(X1,X2)= 0.0602164

R^2_Y,X4|(X1,X2)= 0.00452975

R^2_Y,X5|(X1,X2) = 0.3753792

Interpretation: Because the Value of variable X5 (Number of hospital beds) has the most explanatory power beyond X1 (Total Population) and X2 (Total personal income (millions of dollars))

The partial coefficient of determination is about 0.3753792. It means that adding the number of hospital beds (X5) reduces the remaining variation in the Number of active physicians by about 37.53792%, by the total population (X1) and total personal income (X2) have being accounted for. And in the land area (X3) and the percent of population aged 65 or older (X4), only reduced by the remaining variation was reduced by 6.02164% and 0.452975.

2. **Partial F-test for Number of hospital beds (X5)**

   Based on the last part, we know that the number of hospital beds has the largest explanatory power. So we use this partial F-test to determine whether adding X5 significantly improves the model.

Null hypothesis: $\beta 5 = 0$
Alternative hypothesis: $\beta 5 \neq 0$

Significance level (a): 0.01
Test Statistics: 262.02
Degree of freedom: (1,436)
P-Value:2.2e-16

Conclusion: Because the P-value is < 0.01, we reject H_0: $\beta_5 = 0$ and conclude that X_5 provides significant additional explanatory power for Y beyond X_1 and X_2. X5 improves the model's ability to predict Y beyond what is already explained by total population (X1) and total personal income (X2).

3. **Partial R^2 for pairs beyond {X1,X2}**

   R^2_Y,(X3,X4)|(X1,X2)= 0.06516955

   R^2_Y,(X3,X5)|(X1,X2)= 0.3878251

   R^2_Y,(X4,X5)|(X1,X2)= 0.3791867

Interpretation: Because the Value of the variable (X3, X5) (Land area and Number of hospital beds) has the most explanatory power beyond (X3, X4) (Land area and Percent of population aged 65 or older) and (X4, X5) (Total personal income (millions of dollars))

The partial coefficient of determination is about 0.3878251. It means that adding both the Land area (X3) and the number of hospital beds (X5) reduces the remaining variation in the Number of active physicians by about 38.78251%. And in the land area (X3) and the percentage of the population aged 65 and older (X4) for the remaining variation is 6.516955%, the percentage of the population aged 65 and older (X4) and the number of hospital beds (X5) for the remaining variation is 37.91867%.

4. **Partial F-test for Land area and Number of hospital beds (X3, X5)**

Based on the last part, we know that the number of hospital beds has the largest explanatory power. So we use this partial F-test to determine whether adding X5 significantly improves the model.

Null hypothesis: $\beta 3 = \beta 5 = 0$

Alternative hypothesis: At least one $\beta 3$, $\beta 5 \neq 0$

Significance level (a): 0.01

Test Statistics: 137.79

Degree of freedom: (2,435)

P-Value:2.2e-16

Conclusion: Because the P-value < 0.01, we reject H_0: $\beta 3 = \beta 5 = 0$. So we conclude that adding Land Area (X3) and the Number of Hospital Beds (X5) simultaneously to the model improves the model's ability to predict Y beyond what is already explained by total population (X1) and total personal income (X2).

Part III — Discussion

Which predictors appear most important, and why?

Total population (X1), total personal income (X2), and the number of hospital beds (X5) are statistically significant predictors of active physicians in both models. This observation is supported by both correlation and regression results. Results in Model A indicate extremely strong correlations between Y and population and between Y and income (approx. 0.94 and 0.95 respectively); meanwhile, the correlation between Y and land area (X3) is practically zero. These much larger positive correlations show that population and income have a good linear relationship with physician number alone and that land area has almost no linear predictive power. In Part II, the partial coefficient of determination results show that X5 gives the greatest additional explanatory power beyond population and income, with a partial R^2 value of ~0.375. The significance of hospital beds lies in the fact that X5 has the highest partial R^2 across all additional variables, and its partial F-test is very strong ($F = 262.02$, df = 1, 436, $p < 0.0001$), indicating a large amount of the residual variation after controlling population and income is dropped by including X5. Overall, this means that once size and income are factored in, counties that have more medical infrastructure tend to have more physicians.

Did interaction terms improve predictive performance or interpretation?

By including the two-factor interactions between the predictors in each model and re-fitting them, the overall R^2 has also risen. This indicates that the interaction terms offer extra explanatory information above and beyond the main effects alone. The increased R^2 after the influence of interaction terms indicates that these effects carry over significant extra variability in Y not captured by the main-effect model. Although the statistical fit improved, the main conclusions did not change. Total income and population remain the main predictors. Hospital beds provide the largest additional explanatory power in Part II. However, the interaction coefficients make the model more complex and, when viewed practically, harder to interpret. Consequently, interactions enhanced the predictive performance of the model, but they did not change the interpretation of which variables are the most important.

Key statistical concepts used

Several fundamental statistical principles were utilized in this work. Correlation and scatterplot matrices were used to investigate linear relationships and to look for potential multicollinearity. For instance, the correlation between population (X1) and income (X2) is about 0.99, indicating serious multicollinearity, and hence, explains why their individual regression coefficients in Model A can be quite difficult to

interpret. Multiple linear regression models were fitted, and the estimated coefficients were interpreted in the appropriate context. The model comparison was performed using R^2 and adjusted R^2. Residual plots and normal Q–Q plots were applied to assess model adequacy. The additional explanatory power of the variables beyond the baseline model with X1 and X2 was analysed using partial coefficients of determination. Partial F-tests indicated whether the additional variables significantly reduced the remaining variance, and thus, the corresponding conclusions were based on their p-values.

Diagnostics, limitations, and suggestions for improvement

Diagnostics for the models indicate they are fairly good, but the distributions of several predictors have strong right-skewness, including population, total income, land area, and population density. The histograms for these variables indicate that most counties are of small or moderate size, whereas a few extremely large counties produce strong right tails, validating the existence of large outliers. The residual plots show that, for most counties, the estimates by the models are valid, but some extreme counties can result in a mild heteroscedasticity. The combination of population and income in Model A is such that the results are extremely correlated that it creates multicollinearity (i.e., we cannot evaluate them separately by their coefficients). Possible ways to make it better are to transform the skewed predictors (like if we take logarithms of population or income), down-weight outliers, if needed explore nonlinear relationships.

Practical interpretation of findings

The results are consistent with real-world expectations. Counties that have a larger population and a higher total income generally support more physicians, and counties that have more hospital beds also have more physicians, controlling for both population and income. These results highlight the overall significance of population size, economic resources, and medical infrastructure in deciding physician availability. These regression models set out to quantitatively compare the strength of these relationships. At the same time, the shortcomings observed in the diagnostics also show that real-world data tend to be skewed and imbalanced and that model selection must balance statistical fit, interpretability, and practical relevance.

Code Appendix:

```
#Import Data CDI.txt
install.packages("tidyverse")
library(tidyverse)
CDI <- read_table(file.choose(), col_names = FALSE)
```

#Part I Question 1

```
colnames(CDI) = c(
  "ID", "County", "State", "Area", "Population",
  "Per_18_34", "Per_65plus", "MDs", "Beds",
  "Crimes", "HS", "BA", "Poverty", "Unemp",
  "PCI", "TPI", "Region"
)

data = CDI[,c(8,5,16,4,7,9)]
colnames(data) = c("Y", "X1", "X2", "X3", "X4", "X5")
Density = data$X1 / data$X3
summary(data)

hist(data$X1, main="Population")
hist(data$X3, main="Land Area")
```

```
hist(data$X2, main="Total Income")
hist(data$X4, main="Percent 65+")
hist(Density, main="Population Density")
#Part I Question 2
data$Density = Density
pairs(data[, c("Y", "X1", "X3", "X2")],
     main = "Scatterplot Matrix - Model A")

pairs(data[, c("Y", "Density", "X4", "X2")],
     main = "Scatterplot Matrix - Model B")

cor_matrix_A = cor(data[, c("Y", "X1", "X3", "X2")])
cor_matrix_A

cor_matrix_B = cor(data[, c("Y", "Density", "X4", "X2")])
cor_matrix_B

#Part I Question 3
modelA = lm(Y ~ X1 + X3 + X2, data=data)
modelB = lm(Y ~ Density + X4 + X2, data=data)

summary(modelA)
summary(modelB)
#Part I Question 5

par(mfrow = c(1, 3))

  plot(data$X1, resid(modelA),
            xlab = "X1: Population",
            ylab = "Residuals",
            main = "Model A: Residuals vs X1")
 abline(h = 0, lty = 2)

  plot(data$X3, resid(modelA),
            xlab = "X3: Land Area",
            ylab = "Residuals",
            main = "Model A: Residuals vs X3")
 abline(h = 0, lty = 2)

 plot(data$X2, resid(modelA),
            xlab = "X2: Total Income",
            ylab = "Residuals",
            main = "Model A: Residuals vs X2")
 abline(h = 0, lty = 2)

  par(mfrow = c(1, 3))

  plot(data$Density, resid(modelB),
            xlab = "Density",
            ylab = "Residuals",
            main = "Model B: Residuals vs Density")
```

```
abline(h = 0, lty = 2)

  plot(data$X4, resid(modelB),
          xlab = "X4: Percent 65+",
          ylab = "Residuals",
          main = "Model B: Residuals vs X4")
abline(h = 0, lty = 2)

  plot(data$X2, resid(modelB),
          xlab = "X2: Total Income",
          ylab = "Residuals",
          main = "Model B: Residuals vs X2")
abline(h = 0, lty = 2)
```

#Part I Question 6

```
modelA_int = lm(Y ~ X1*X2 + X1*X3 + X2*X3, data=data)
summary(modelA_int)
modelB_int = lm(Y ~ Density*X4 + Density*X2 + X4*X2, data=data)
summary(modelB_int)
```

#Part II Question 1

```
model_base = lm(Y ~ X1 + X2, data=data)
model_X3 = lm(Y ~ X1 + X2 + X3, data=data)
model_X4 = lm(Y ~ X1 + X2 + X4, data=data)
model_X5 = lm(Y ~ X1 + X2 + X5, data=data)

sse_base   = deviance(model_base)
sse_X3 = deviance(model_X3)
sse_X4 = deviance(model_X4)
sse_X5 = deviance(model_X5)

r2_X3_X1X2 = ((sse_base - sse_X3)/ sse_base)
r2_X4_X1X2 = ((sse_base - sse_X4) / sse_base)
r2_X5_X1X2 = ((sse_base - sse_X5) / sse_base)

r2_X3_X1X2
r2_X4_X1X2
r2_X5_X1X2
```

#Part II Question 2

```
anova(model_base, model_X5)
```

#Part II Question 3

```
model_X3X4 = lm(Y ~ X1 + X2 + X3 + X4, data=data)
model_X3X5 = lm(Y ~ X1 + X2 + X3 + X5, data=data)
model_X4X5 = lm(Y ~ X1 + X2 + X4 + X5, data=data)

sse_X3X4 = deviance(model_X3X4)
sse_X3X5 = deviance(model_X3X5)
sse_X4X5 = deviance(model_X4X5)

r2_X3X4_X1X2 = (sse_base - sse_X3X4) / sse_base
r2_X3X5_X1X2 = (sse_base - sse_X3X5) / sse_base
r2_X4X5_X1X2 = (sse_base - sse_X4X5) / sse_base
```

```
r2_X3X4_X1X2
r2_X3X5_X1X2
r2_X4X5_X1X2
```

#Part II Question 4

```
anova(model_base, model_X3X5)
```